NGO CHI NGHIA

AI Engineer | RAG/LLM Systems | Computer Vision

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SUMMARY

Information Technology graduate specializing in AI Engineering with expertise in Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and Computer Vision. Experienced in designing, developing, and deploying AI applications using FastAPI, LangChain, LangGraph, Docker, and modern LLM APIs. Passionate about building production-ready AI systems that solve real-world problems.

EDUCATION

Nam Can Tho University (NCTU) — Can Tho, Vietnam
Bachelor of Information Technology — *Oct. 2022 – Expected Aug. 2026*

- **GPA:** 3.44/4.0

TECHNICAL SKILLS

Languages: Python, SQL, C/C++, TypeScript/JavaScript
LLM & Retrieval: LangChain, LangGraph, Prompt Engineering, ChromaDB, BM25, RRF, multilingual-E5
Machine Learning: PyTorch, TensorFlow, Scikit-learn, YOLOv8, OpenCV, HuggingFace
Backend: FastAPI, REST API
DevOps: Docker, Docker Compose, Nginx, Linux, Git
Frontend: React

EXPERIENCE

Tien Phong Engineering Technology Co., Ltd. — Can Tho, Vietnam
AI Engineer Intern — *Mar. 2026 – Jul. 2026*

- Developed RESTful AI services using FastAPI for LLM-powered applications.
- Designed and optimized Retrieval-Augmented Generation (RAG) pipelines using ChromaDB and multilingual-E5 embeddings.
- Integrated OpenAI, Gemini, and Groq APIs into AI workflows.
- Containerized and deployed AI services using Docker on Ubuntu Linux.
- Collaborated with software engineers to integrate AI models into production systems.

SELECTED PROJECTS

Energy-Based Retrieval Clustering RAG for Economy QA | *Python, LangChain, E5, KMeans, FastAPI* — Jan. 2026

- Proposed and implemented ERC-RAG, a cluster-level retrieval method using KMeans and energy distance over multilingual-E5 embeddings.
- Indexed VEID into 120,165 ChromaDB vectors and evaluated on 1,000 VNEIQAD questions using BLEU-2, ROUGE-2, MRR, Hit@5 and NDCG@5.
- Achieved BLEU-2 0.71, ROUGE-2 F1 0.74, Cosine 0.96, MRR 0.68, Hit@5 0.70 and NDCG@5 0.87 in the multi-query setting.

PolyAqua Agent | *FastAPI, YOLOv8, ChromaDB, BM25, RRF, OpenAI, Pytest* — 2026

- Built an AI assistant for shrimp-crab polyculture combining disease image classification, aquaculture calculators and RAG-based treatment recommendations.
- Trained YOLOv8n-cls.
- Ingested 13 aquaculture technical PDFs into ChromaDB with 4,593 vectors and implemented E5 + BM25 hybrid retrieval with RRF fusion.

Vietnamese Legal Document RAG Web App | *FastAPI, LangGraph, ChromaDB, React, Docker* — Jan. 2026

- Deployed lawchat.me, a Vietnamese legal QA web app over 6 IT/cybersecurity-related legal documents.
- Implemented hybrid retrieval with ChromaDB and BM25, plus LangGraph workflow and grounded prompting for source-based answers.
- Containerized backend/frontend with Docker Compose and Nginx on Ubuntu server; built React chat UI with Markdown rendering.

AI-Generated Image Detection (CIFAKE) | *TensorFlow, Keras, CNN* Dec. 2025

- Built and trained a Convolutional Neural Network (CNN) to classify Real vs. AI-generated images, achieving an accuracy of $\sim$0.9261.
- Applied Data Augmentation techniques (rotation, zoom, flip) to prevent overfitting and optimized hyperparameters to achieve best validation performance.

RESEARCH & COMPETITIONS

Scientific Research Publications 2026

Co-authored 2 research papers on RAG architecture optimization:

- *"ERC-RAG: An Energy-based Retrieval Clustering for RAG in Vietnamese Economy Chatbots"* - Proposed a two-stage retrieval architecture for the VEID benchmark.
- *"Multi-Query ERC-RAG: An Energy-based Retrieval Clustering Framework for Retrieval-Augmented Generation"* - Demonstrated consistent improvements in retrieval effectiveness and answer quality.

FPTU Can Tho Olympic AI 2026 Can Tho, Vietnam

Finalist Certificate

- Competed and reached the final round in the university-level Artificial Intelligence competition.

LANGUAGES

Language: English (VSTEP B1)